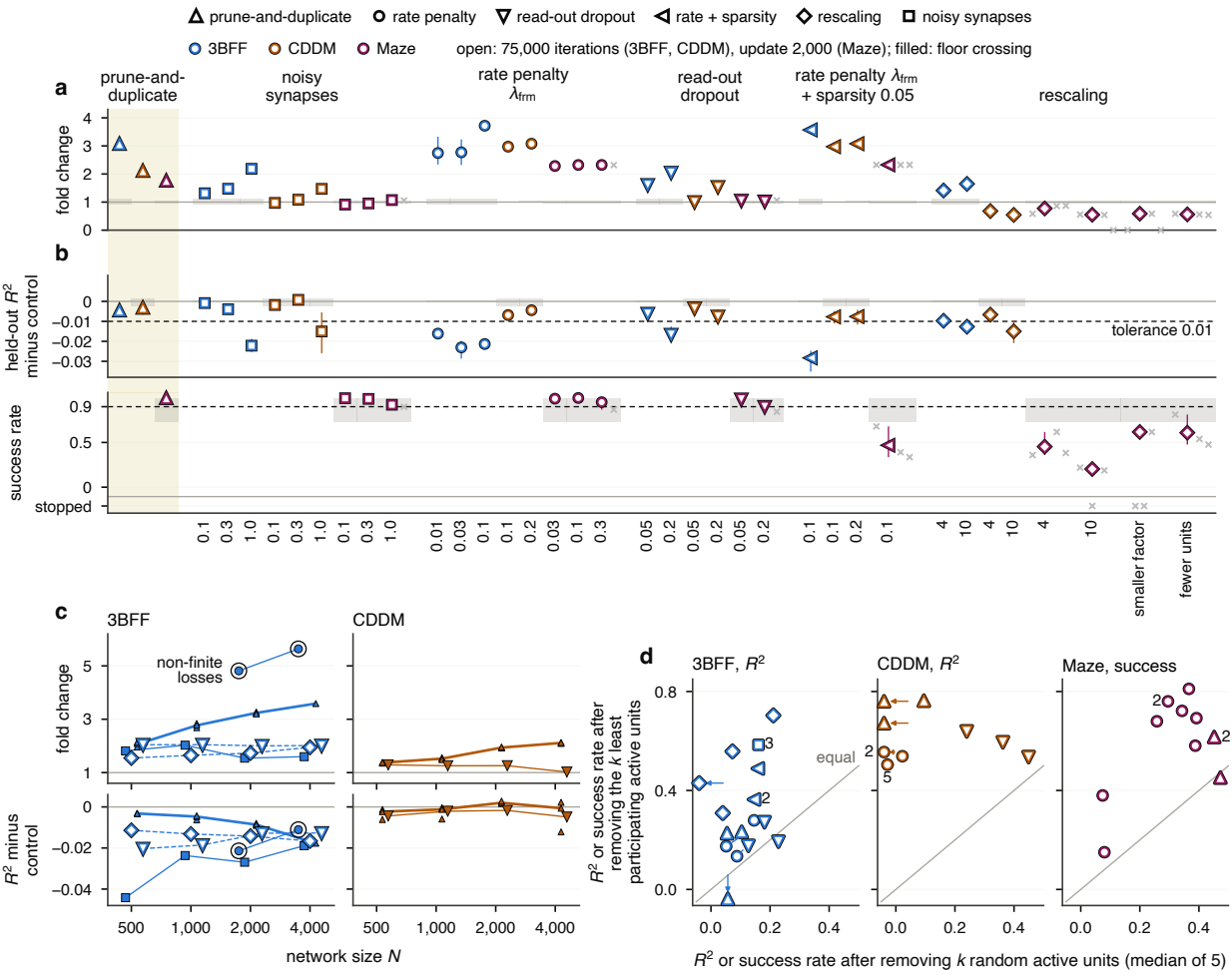
prune-and-duplicate
rate penalty
read-out dropout
rate + sparsity
rescaling
noisy synapses
3BFF
CDDM
Maze
open: 75,000 iterations (3BFF, CDDM), update 2,000 (Maze); filled: floor crossing
a
prune-and-duplicate
noisy synapses
rate penalty λ_{frm}
read-out dropout
rate penalty λ_{frm} + sparsity 0.05
rescaling
fold change
b
held-out R^2 minus control
tolerance 0.01
success rate
stopped
0.1 0.3 1.0 0.1 0.3 1.0 0.1 0.3 1.0 0.01 0.03 0.1 0.1 0.2 0.03 0.1 0.3 0.05 0.2 0.05 0.2 0.05 0.2 0.1 0.1 0.2 0.1 4 10 4 10 4 10 smaller factor fewer units
c
3BFF
CDDM
non-finite losses
fold change
R^2 minus control
500 1,000 2,000 4,000
network size N
d
3BFF, R^2
CDDM, R^2
Maze, success
equal
R^2 or success rate after removing the k least participating active units
R^2 or success rate after removing k random active units (median of 5)